

Supplementary Results Analysis: Comprehensive Evaluation of Feature Engineering and Classification Pipelines Across Three IDS Datasets

S1. Overview of the Experimental Analysis

This supplementary analysis provides an extended evaluation of the proposed pipeline-centric intrusion detection framework across three heterogeneous benchmark datasets: **AWID**, **UNSW-NB15**, and **CSE-CIC-IDS2018**. While the main manuscript focuses on the dominant trends and the highest-performing configurations, this analysis investigates the complete experimental landscape by examining:

1. The effect of feature-selection strategies (FS)
2. The contribution of feature-extraction approaches (FE)
3. The interaction between FS and FE transformations
4. The influence of classifier families
5. The consistency of performance trends across datasets
6. The stability of improvements under repeated validation

The objective is to determine whether observed performance improvements originate from isolated model advantages or from systematic improvements in the transformation of high-dimensional network measurements.

S2. Dataset-Level Performance Characteristics

The three datasets exhibit different levels of classification difficulty, which is reflected in the achieved Macro-F1 values.

Among the evaluated datasets, **AWID achieves the highest overall performance**, followed by **UNSW-NB15**, while **CSE-CIC-IDS2018 presents the most challenging classification environment**.

The best-performing pipelines achieve approximately:

Dataset	Best Pipeline	Macro-F1
AWID	Chi2 → LDA → STACK	0.8361
UNSW-NB15	Chi2 → LDA → STACK	0.7924
CSE-CIC-IDS2018	Chi2 → LDA → STACK	0.7518

The gradual decrease in performance from AWID to UNSW-NB15 and CSE-CIC-IDS2018 indicates increasing complexity caused by differences in:

- traffic diversity,
- attack-category distribution,
- class overlap,
- feature heterogeneity,
- and network behavior variability.

Importantly, the same optimal processing pattern emerges across all datasets, demonstrating that the observed improvements are not dataset-specific artifacts but rather represent a consistent property of the proposed pipeline design.

S3. Impact of Feature Selection Strategies

S3.1 Aggregated Feature Selection Performance

The feature-selection analysis evaluates each FS method across all extraction methods and classifiers.

The results show a clear distinction between supervised and unsupervised selection strategies.

The overall ranking is:

$$ANOVA > Chi2 > VT$$

for all evaluated datasets.

AWID:

FS Method	Macro-F1
ANOVA	0.7629
Chi2	0.7514
VT	0.7166

UNSW-NB15

FS Method	Macro-F1
ANOVA	0.7392
Chi2	0.7325
VT	0.7084

CSE-CIC-IDS2018

FS Method	Macro-F1
ANOVA	0.7018
Chi2	0.6954
VT	0.6729

S3.2 Interpretation of Feature Selection Behavior

The superiority of ANOVA and Chi2 demonstrates the importance of preserving class-discriminative variables before constructing the final representation.

ANOVA evaluates the statistical difference between classes by measuring the relationship between:

- inter-class variance,
- intra-class variance.

Therefore, features selected by ANOVA are more likely to contain information relevant to distinguishing benign traffic from multiple attack categories.

Chi2 provides slightly lower average performance but achieves the strongest individual pipelines when combined with LDA and stacking.

This difference is important:

- **ANOVA provides the most consistent feature subset**
- **Chi2 provides the strongest task-specific feature subset when combined with appropriate downstream processing**

Therefore, feature-selection effectiveness cannot be interpreted independently. The usefulness of selected variables depends on how they are transformed and classified afterward.

S3.3 Limitation of Variance Threshold Selection

Variance Threshold consistently produces weaker results.

This behavior is expected because VT evaluates only statistical dispersion and does not consider class information.

A feature may have:

- low variance,
- but high discriminative value for a minority attack class.

Therefore, variance-based filtering can remove features that are important for IDS discrimination.

The higher variability observed with VT further indicates that its selected feature subsets are more sensitive to downstream classifier and representation choices.

S4. Feature Extraction Analysis

S4.1 Overall Feature Extraction Ranking

Across all datasets, the feature-extraction ranking remains consistent:

$$LDA > ICA > PCA \approx SVD$$

This demonstrates that the method used to construct the final feature representation has a major influence on IDS effectiveness.

S4.2 Linear Discriminant Analysis Dominance

LDA achieves the highest average performance:

Dataset	LDA Macro-F1
AWID	0.7617
UNSW-NB15	0.7395
CSE-CIC-IDS2018	0.7043

The advantage of LDA originates from its supervised projection mechanism.

Unlike PCA and SVD, which maximize variance preservation, LDA explicitly optimizes:

- maximum separation between attack classes,
- minimum variation within the same class.

This property directly matches the objective of intrusion detection, where the goal is not simply compression but discrimination between multiple traffic categories.

S4.3 PCA and SVD Behavior

PCA and SVD provide comparable results because both rely on low-rank matrix decomposition.

However, their weaker performance indicates that maximum variance preservation does not necessarily correspond to maximum classification relevance.

High-variance traffic characteristics may represent:

- dominant benign traffic patterns,
- frequent attack behaviors,
- or common network characteristics.

Meanwhile, minority attack information may exist in lower-variance directions.

Therefore, unsupervised compression can unintentionally remove discriminative information.

S5. Classifier-Level Analysis

S5.1 Ensemble Models Provide the Strongest Performance

The classifier comparison demonstrates a clear hierarchy:

STACK > RF ≈ ET > Boosting > Classical Models

The stacking classifier achieves:

Dataset	Macro-F1
AWID	0.8589
UNSW-NB15	0.8247
CSE-CIC-IDS2018	0.8137

S5.2 Advantage of Stacking Architecture

The superior performance of STACK originates from combining multiple learning perspectives.

The architecture integrates:

- Random Forest,
- XGBoost,
- LightGBM,

with a meta-classifier.

Each base learner captures different properties:

Model	Strength
RF	nonlinear feature interactions

XGB	sequential error correction
LGBM	efficient gradient optimization

The meta-classifier combines these complementary predictions, producing a more robust decision boundary.

This explains why stacking consistently exceeds individual ensemble models.

S5.3 Individual Ensemble Models

Random Forest and Extra Trees provide consistently strong performance.

For example:

- RF reaches 0.8348 Macro-F1 on AWID,
- 0.8012 on UNSW-NB15,
- 0.7888 on CSE-CIC-IDS2018.

These results confirm that tree-based ensembles are highly suitable for IDS because network traffic commonly contains:

- nonlinear relationships,
- feature interactions,
- heterogeneous attack signatures.

S6. Ablation Study: Contribution of Feature Extraction

The ablation analysis provides the strongest evidence regarding the role of feature extraction.

The comparison evaluates:

- FS-only against $FS \rightarrow FE$

S6.1 General Observation

Feature extraction is not universally beneficial.

Its impact depends strongly on the extraction method.

The results show:

- LDA consistently improves performance.
- PCA and SVD frequently reduce performance.
- ICA provides moderate improvements in some cases.

This confirms that dimensionality reduction should not be considered inherently advantageous.

The transformation must preserve decision-relevant information.

S6.2 Strongest Observed Improvements

The largest gains occur with supervised extraction.

Examples:

Chi2 → LDA

UNSW-NB15: +0.1397

CSE-CIC-IDS2018: +0.1394

These are substantial improvements compared with unsupervised approaches.

This demonstrates that feature selection followed by supervised representation learning creates a more separable feature space.

S6.3 Dataset Complexity and Feature Extraction Benefit

A major observation is that FE becomes more valuable as dataset complexity increases.

For AWID:

- LDA provides improvement,
- but the gain is smaller.

For UNSW-NB15 and CSE-CIC-IDS2018:

- LDA produces much larger improvements.

This suggests that complex datasets contain more redundant and overlapping structures, making supervised projection increasingly valuable.

S7. Cross-Dataset Consistency Analysis

One of the strongest findings is the stability of the ranking patterns.

Across all datasets:

Feature Selection

$$ANOVA > Chi2 > VT$$

Feature Extraction

$$LDA > ICA > PCA \approx SVD$$

Classifier

$$STACK > RF/ET > Boosting > Classical$$

The preservation of these rankings across three different IDS environments demonstrates that the proposed conclusions are not dependent on a single benchmark.

S8. Main Experimental Insights

The comprehensive evaluation leads to several important conclusions.

Insight 1: Representation quality is as important as classifier complexity

High-performing classifiers alone cannot guarantee strong IDS performance.

The classifier benefits directly from the quality of the transformed feature space.

Insight 2: Supervised transformations are more suitable for IDS

Methods incorporating class information consistently outperform unsupervised approaches.

This applies to both:

- feature selection,
- feature extraction.

Insight 3: Feature engineering should be evaluated as a pipeline

The best individual components do not always create the best complete system.

For example:

- ANOVA gives the best average FS performance.
- Chi2 creates the strongest end-to-end pipeline.

Therefore, preprocessing components must be evaluated jointly.

Insight 4: Stacking provides robust generalization

The consistent superiority of STACK indicates that hybrid ensemble learning is particularly effective for heterogeneous IoT intrusion detection.

S9. Overall Supplementary Conclusion

The extended experimental analysis confirms that IDS performance emerges from the interaction between measurement transformation and classification strategy.

The strongest and most transferable configuration is:

$$\boxed{Chi2 \rightarrow LDA \rightarrow STACK}$$

because it combines:

1. supervised feature selection,
2. discriminative latent representation learning,
3. ensemble decision modeling.

The results demonstrate that improving intrusion detection performance requires moving beyond isolated algorithm selection toward a complete pipeline-centric design philosophy, where feature processing and classification are optimized as interconnected stages of a measurement-data processing system.